

Proof of Concept

Adaptive THz Access Point Placement Using APSO with 4-Tier Hierarchical Escalation

IIT Bhilai

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1. Problem Statement

- THz (300 GHz) signals suffer from severe path loss, molecular absorption, and blockage.
- Fixed access points cannot adapt to changing user/crowd distributions.
- Electronic beamforming alone fails when all line-of-sight paths are blocked by walls or human bodies.
- Baseline coverage drops to 60–75% in dynamic indoor environments—far below the 90% target.

2. Proposed Solution

- 14 THz routers mounted on **motorized ceiling rails** for physical repositioning.
- **APSO (Adaptive Particle Swarm Optimization)** computes optimal router positions in real-time.
- **4-Tier Escalation Hierarchy** to minimize unnecessary mechanical movement:
 - Tier 1–2: Electronic adaptation (beam azimuth ϕ , beamwidth α).
 - Tier 3: Vertical height adjustment (Z -axis, 2.0 m–3.2 m).
 - Tier 4: Full 3D spatial repositioning (X, Y, Z).
- **Hungarian Algorithm** for minimum-displacement router-to-target assignment.

3. Simulation Setup

- **Room:** 32 m $\times$ 20 m $\times$ 3 m with realistic wall layout (concrete, wood, glass).
- **Routers:** 14 THz APs at 300 GHz, 20 dBm Tx power, 30 dBi max antenna gain.
- **Channel Model:** FSPL + molecular absorption ($k_f = 0.05$) + 3D ray-wall intersection + human body blockage (20 dB).
- **Material Losses:** Concrete 30 dB, Wood 20 dB, Glass 15 dB.
- **APSO Config:** 200 particles, 70 dimensions (14 $\times$ 5 params), 200 iterations, adaptive inertia (0.9 $\rightarrow$ 0.4).
- **Fitness Function:** Density-weighted coverage (0.6) + cell coverage (0.4) + target bonus – movement penalty – anti-clustering penalty.

- **Time Steps:** 3 dynamic crowd scenarios (t_0, t_1, t_2) with shifting Gaussian user distributions.

4. Results & Evidence

- **Coverage Improvement:**
 - Before APSO: $\sim 60\text{--}75\%$.
 - After APSO: $\geq 90\%$ (target met across all 3 time steps).
- **Optimization Speed:** < 2.5 seconds per cycle (real-time feasible).
- **Total Router Displacement:** $< 5\text{--}12$ meters combined across all 14 routers.
- **Reachability Gain:** $+15\text{--}35\%$ coverage increase per meter of movement.
- **Dead Zone Reduction:** $\sim 90\%$ elimination of poor-SINR areas.
- **SINR Heatmaps:** Clear visual improvement—dead zones before $\rightarrow$ uniform coverage after.
- Coverage maintained consistently across all dynamic crowd shifts.

5. Novelty & Differentiation

- No existing system combines physical AP movement + THz-specific 3D propagation + APSO + 4-tier escalation.
- Compared against 6 prior art patents (NYU, Samsung, Peltbeam, DJI)—none offer this combination.
- **4 Novel Pillars:**
 - 4-Tier escalation hierarchy (electronic first, physical last).
 - APSO with density-weighted fitness and anti-clustering constraints.
 - Hungarian minimum-displacement assignment.
 - Material-aware and height-aware 3D THz propagation model.
- Patent invention disclosure filed (60 claims, 6 target jurisdictions).

6. Working Demo

- **Interactive Web Dashboard** (React + Vite + FastAPI): tune parameters, run optimization, see SINR heatmap update live.
- **CHSR Hierarchy Visualizer:** animated 4-tier communication flow.
- **Python Simulation:** generates all plots (SINR heatmaps, AP placement, density maps, coverage charts) in seconds.
- Numba JIT-compiled physics for real-time performance on commodity hardware.
- **Video Demo:** <https://youtu.be/demo-placeholder-link>

7. Conclusion

- The concept is **validated through simulation** with quantitative and visual evidence.
- Coverage target of $\geq 90\%$ is **consistently achieved** across dynamic scenarios.
- Optimization is **real-time feasible** (< 2.5 s) with minimal physical movement.
- Solution is **technically feasible**—ceiling rails, THz hardware, and APSO all exist today.
- **IP protected**—patent disclosure filed at IIT Bhilai.